2. **Install the JDK:**
   - Run the downloaded installer and follow the instructions.
   - Set the JAVA_HOME environment variable to the JDK installation directory.
3. **Verify the Installation:**
   - Open your terminal or command prompt.
   - Type java -version and press Enter. You should see the installed Java version.

Installing an Integrated Development Environment (IDE):

1. **Download an IDE:**
   - Popular IDEs for Java include IntelliJ IDEA, Eclipse, and NetBeans.
   - IntelliJ IDEA: Download from JetBrains (*https://www.jetbrains.com/idea/download/).*
   - Eclipse: Download from Eclipse.
   - NetBeans: Download from NetBeans (*https://netbeans.apache.org/download/index.html).*
2. **Install the IDE:**
   - Run the downloaded installer and follow the instructions.
   - Configure the IDE to use the installed JDK.

Basic Java Syntax

Understanding the basic syntax of Java is essential for writing your first program. Here are some fundamental concepts and constructs:

Variables and Data Types:

Java supports various data types, including integers, floats, characters, and booleans. Variables must be declared with a specific type.

```
// Integer
int age = 25;

// Float
float height = 5.9f;

// Character
char initial = 'A';

// Boolean
boolean isStudent = true;
```

Comments:

Comments in Java can be single-line, starting with //, or multi-line, enclosed in /* */.

```
// This is a single-line comment

/*
This is a
multi-line comment
*/
```

Control Structures:

Java supports control structures like conditionals and loops.

```
// Conditional
if (age >= 18) {
    System.out.println("You are an adult.");
} else {
    System.out.println("You are a minor.");
}

// Loop
for (int i = 0; i < 5; i++) {
    System.out.println(i);
}
```

Functions (Methods):

Functions in Java are called methods. They are defined within a class and can return a value.

```
public class Main {
    public static void main(String[] args) {
        System.out.println(greet("Alice"));
    }

    public static String greet(String
name) {
        return "Hello, " + name + "!";
    }
}
```

Object-Oriented Programming:

Java is an object-oriented language, meaning it revolves around objects and classes.

```
// Define a class
public class Person {
    // Attributes
    String name;
    int age;

    // Constructor
    public Person(String name, int age) {
```